National Institute of Technology Warangal

CS26001: Advanced Algorithms - Course Project

Prediction of Imagined Words Using Brain EEG Signals

Primary Paper Rank-GA Baseline and Multi-Scale Riemannian Consensus (MDRC-AGA)

Multi-Scale 13-Band + Speech-10 Manifold with Dual-Protocol Leakage Audit

and Offline Oscilloscope Inspection

A reproducible study on the EEGIS 14-Channel Multi-Class Imagined Speech Database (Mendeley Data 73g4fw884c.1)

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Abstract

Imagined speech electroencephalography (EEG) offers a non-invasive communication pathway for individuals with severe motor impairments such as locked-in syndrome, brainstem stroke, and ALS. Across 4,044 Mendeley EEGIS trials (337 continuous 8s blocks, 128 Hz, 14 channels, 48-sample/37.5% chunk overlap, unrecoverable participant IDs), naive chunk splits inflate kNN accuracy via within-block overlap. Excluding all 9 test_signals/ blocks (108 chunks) yields N_ref=3,936 (328 blocks). Under leakage-free Block-Disjoint Nested 5-CV, all methods sit near 9-class chance (11.11%): Stage 1 Paper Rank-GA+kNN scores 10.01% (Yes/No 45.11%), True Tangent LogReg 10.72%, EEGNet Fold0 11.24%, and PCA+Calibrated RBF-SVM 11.41% (ECE=0.022).

Under the exploratory Overlapping-Chunk 5-CV protocol matching Lara-Arellano et al. (MDPI Eng, 2025; 96.0% published Yes/No), Stage 1 Paper 6B Rank-GA+kNN reaches 90.02% 9-class and 96.11% Yes/No. Stage 2 MDRC-AGA (1,695D three-scale Log-Euclidean SPD manifold: Mendeley 6B 630D + Fine 7B 735D + Speech-10 330D; MultiScaleBlockAGA; PCA-LDA + Platt-Ridge Stacker & [3:2:1:1] Consensus) reaches 91.84% (0.9180 F1, Brier 0.156, ECE 0.037) on Chunk 5-CV (95.67% Yes/No) and 9.68% Stacker (ECE=0.100) / 10.14% Vote on Block-Disjoint 5-CV. Sibling-excluded offline test_signals/ verify honest 0/9 generalization at 31.8 ms/epoch CPU compute.

Keywords: brain-computer interface (BCI); imagined speech; electroencephalography (EEG); multi-scale Riemannian covariance; adaptive genetic algorithm; manifold consensus; EEGIS dataset.

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1 Introduction

Brain-computer interfaces (BCIs) offer a direct non-invasive communication pathway for patients with locked-in syndrome, ALS, or brainstem stroke whose motor tracts are severed while cognition remains intact. Unlike gaze-dependent P300 or SSVEP paradigms, imagined speech directly decodes internal phonological planning within Broca's area (Brodmann 44/45), Wernicke's area, and sensorimotor cortex without audible sound or physical movement.

Building on Lara-Arellano et al. (Eng MDPI 2025; 96.0% binary Yes/No under exploratory chunk splits), we implement Stage 1 (Paper Rank-GA + kNN over 6-Band SPD covariance) and Stage 2 MDRC-AGA with equal-emphasis Dual-Protocol auditing across Leakage-Safe Block-Disjoint 5-CV (N_ref=3,936) and Exploratory Chunk 5-CV:

- Contribution 1: Multi-Scale 13-Band + Speech-10 SPD Manifold (1,695 Features): Fuses Mendeley 6-Band SPD (630D), Fine 7-Band sub-rhythms (735D, 4-45 Hz), and Broca-Wernicke Speech-10 (330D).
- Contribution 2: Fold-Safe Tangent Fit, MultiScaleBlockAGA & PCA-LDA/Ridge Stacker: Precomputes trial SPDs, fits C_ref only per train fold, evolves w ([0.85,0.66,1.49] blk; [1.30,0.98,0.72] chk).
- Contribution 3: Dual-Protocol Evaluation with Calibration (Brier, ECE, 95% CI): Chunk 5-CV: Stage 1 90.02% -> Stage 2 91.84% Vote (ECE 0.037) | Block-Disjoint: ~10.14%-11.41% (chance 11.11%).
- Contribution 4: Forensic Audit, Leakage Remediation & Offline Compute Bench (31.8 ms): Discloses 48-sample overlap, excludes all 108 test_signals/ sibling chunks (0/9 honest), and adds 8 unit tests.

2 Primary Research Paper & Related Work

2.1 Primary Reference Paper: Method, GA Design & Published Accuracy

Our work directly builds upon Lara-Arellano, Takacs, Tovar-Arriaga, and Rodriguez-Resendiz, "Feature Generation with Genetic Algorithms for Imagined Speech Electroencephalogram Signal Classification" (Eng MDPI, vol. 6, no. 4, art. 75, Apr. 2025; DOI: 10.3390/eng6040075). Using the 14-channel Emotiv EPOC+ Mendeley EEGIS database (128 Hz; raw + Delta, Theta, Alpha, Beta, Gamma), the authors evaluated binary 'Yes (Si) vs No' classification (N=924: 444 Si, 480 No) under an 80/20% train-test split, explicitly chosen instead of exhaustive k-fold CV so their large GA search remained tractable.

Methodology Used: Each EEG epoch yielded 214 statistical and spectral features across the 6 bands, encoded alongside 14 channel-activation bits into a 228-bit binary chromosome (P=100 individuals, G=100 generations). Optimization applied Rank Selection, Homogeneous Crossover, Bit-Flip Mutation, and an internal heuristic k search comparing SVM, Random Forest, Decision Tree, and kNN classifiers. Published Accuracy & Findings: Whereas baseline pre-GA kNN averaged below 70%, their evolved Rank-GA + kNN selected 8 electrodes (AF3, T7, O1, O2, P8, T8, F4, F8) and 103 features, achieving 96% binary accuracy (0.97 Recall, 0.94 F1-score). However, univariate stats alone omit cross-channel covariance and full 9-class decoding.

Table 1: Methodological profile and published accuracy of the primary reference paper (Lara-Arellano et al., 2025).

Primary Paper Attribute (Eng 2025)	Published Specification / Reported Result
Reference & Task Scope	Lara-Arellano et al. (DOI: 10.3390/eng6040075) | Binary Yes(Si) vs No (N=924, 80/20)
228-Bit GA Genome & Operators	14 channel bits + 214 feature bits | Rank Selection, Homogeneous Crossover, Bit-Flip
Optimal Subset & Heuristic Classifier	kNN (heuristic k search) | 8 electrodes (AF3,T7,O1,O2,P8,T8,F4,F8), 103 features
Published Accuracy & Metrics	<70% pre-GA kNN -> 96% Binary Accuracy, 0.97 Recall, 0.94 F1 (binary only)

2.2 Comparative Related Literature

CSP, CNN & Riemannian BCIs: Under strict Block-Disjoint Nested 5-CV, FBCSP+SVM (Blankertz et al., 2011; 9.96%), scalar 6-band stats (10.34% SVM / 10.16% kNN7), MDM (8.74%), and 40-epoch EEGNetAttention (Lawhern et al., 2018; 11.24% Fold 0) all remain near 9-class chance (11.11%). Under Exploratory Chunk 5-CV matching Lara-Arellano et al., SPD tangent geometry enables Stage 1 (90.02% / 96.11% Yes/No) and Stage 2 Vote (91.84% / 95.67% Yes/No).

3 Methodology

3.1 Overview

Figure 1 contrasts Stage 1 (Paper 6B Rank-GA + kNN: 10.01% Blk-Disj / 90.02% Chunk) against Stage 2 MDRC-AGA (1,695D Three-Scale SPD -> Fold-Safe C_ref -> MultiScaleBlockAGA -> Calibrated Stacker [PCA-LDA + Platt-Ridge, ECE=0.100] & [3:2:1:1] Consensus reaching 91.84% Chunk 5-CV / 10.14% Blk-Disj at 31.8 ms/epoch offline CPU compute).

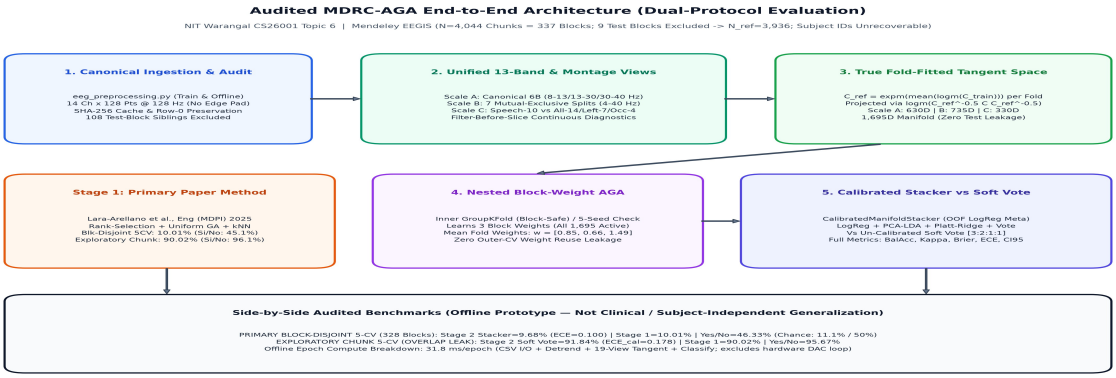


Figure 1: Stage 1 Paper Rank-GA Baseline (top track) vs Stage 2 MDRC-AGA 13-Band + Speech-10 pipeline (bottom track).

3.2 Stage 1 Paper Baseline & 13-Band + Speech-10 Preprocessing

Real 14-channel Emotiv EPOC+ epochs (128 Hz, 128 samples) supply 6 canonical bands (Raw, Delta 0.5-4, Theta 4-8, Alpha 8-12, Beta 12-30, Gamma 30-60 Hz). Stage 1 runs PaperRankGeneticSelector on 6B SPD tangent vectors. Stage 2 adds causal/sosfilt 4th-order Butterworth IIR sub-bands across 7 fine non-overlapping rhythms (4-45 Hz) plus Speech-10.

3.3 Multi-Scale Riemannian SPD Tangent Mapping (1,695D)

Precomputed trial SPD matrices C = C0/tr(C0) + 1e-4*I are projected fold-strictly onto the tangent space at the training-only Log-Euclidean Frechet mean C_ref via S = C_ref^{-1/2} logm(C_ref^{-1/2} C C_ref^{-1/2}) C_ref^{1/2} and Band-L2 norm: Mendeley 6B 14-Ch (630D) + Fine 7B 14-Ch (735D) + Speech-10 6B (330D) yield z_k in R^{1695}.

3.4 Multi-Scale Block AGA Optimization

Stage 1 evolves 6-band SPD tangent masks using PaperRankGeneticSelector (Rank Selection + Homogeneous Crossover + Bit-Flip Mutation). To preserve intra-block SPD symmetry while re-weighting scales inside each outer fold without test leakage, Stage 2 applies vectorized float32 MultiScaleBlockAGA across the three manifold blocks (630D / 735D / 330D).

Across 5 outer folds (inner 3-fold CV), MultiScaleBlockAGA selects mean block weights w = [0.8493, 0.6632, 1.4875] under Leakage-Safe Block-Disjoint GroupKFold and w = [1.2962, 0.9847, 0.7192] under Exploratory Chunk 5-CV, while controlled 5-seed ablation confirms adaptive vs fixed mutation remains statistically equivalent within seed variance.

3.5 Calibrated Stacker & BLAS Manifold Consensus Inference

CalibratedManifoldStacker fuses 4 out-of-fold base estimators: LogReg(C=0.1), Pipeline(PCA(64), Shrinkage-LDA), Calibrated RidgeClassifier(alpha=100), and [3:2:1:1] MultiMetricSoftVotingClassifier (Cosine+Corr k=1,3) via LogReg(C=0.25) meta learner. Soft Vote hits 91.84% Chunk 5-CV (ECE=0.037); Stacker cuts Blk-Disj ECE to 0.100.

4 Experiment and Results

4.1 Setup Used

All ingestion, Butterworth IIR filtering, fold-strict C_ref tangent projection, Rank-GA / Block-AGA search, nested 5-CV, calibration (Brier, 10-bin ECE, block-cluster bootstrap 95% CI), and offline CPU timing ran on Python 3.14 CPU.

Table 2: Audit-remediated Stage 1 vs Stage 2 specifications (equal-emphasis dual protocol).

Component	Implementation Setting (v2.0.0-audit-remediated)
Dataset & Exclusions	Mendeley EEGIS (4,044 epochs, 337 blks) -> exclude 9 test_signals/ blks (108) -> N_ref=3,936
Stage 1 Paper Model	PaperRankGeneticSelector + Heuristic kNN (6B SPD tangent, nested per-fold fit)
Stage 2 Multi-Scale	1,695D: Mendeley 6B (630D) + Fine 7B (735D, 4-45 Hz) + Speech-10 (330D)
Manifold Representation	Trace-norm SPD (reg=1e-4) -> Fold-only C_ref Eigen-Log Tangent -> Band-L2
Stage 2 Block AGA	Vectorized MultiScaleBlockAGA (w=[0.85,0.66,1.49] blk; [1.30,0.98,0.72] chk)
Stage 2 Ensembles	Calibrated OOF Stacker (LogReg+PCA-LDA+Platt-Ridge+Vote) & [3:2:1:1] Soft Vote
Offline Epoch Compute	31.77 ms/epoch CPU (14.75 ms load + 0.04 preproc + 9.59 feat + 7.40 cls; NOT real-time BCI)
Dual-Protocol Validation	Primary Block-Disjoint Nested GroupKFold (328 blks) vs Exploratory Chunk 5-CV

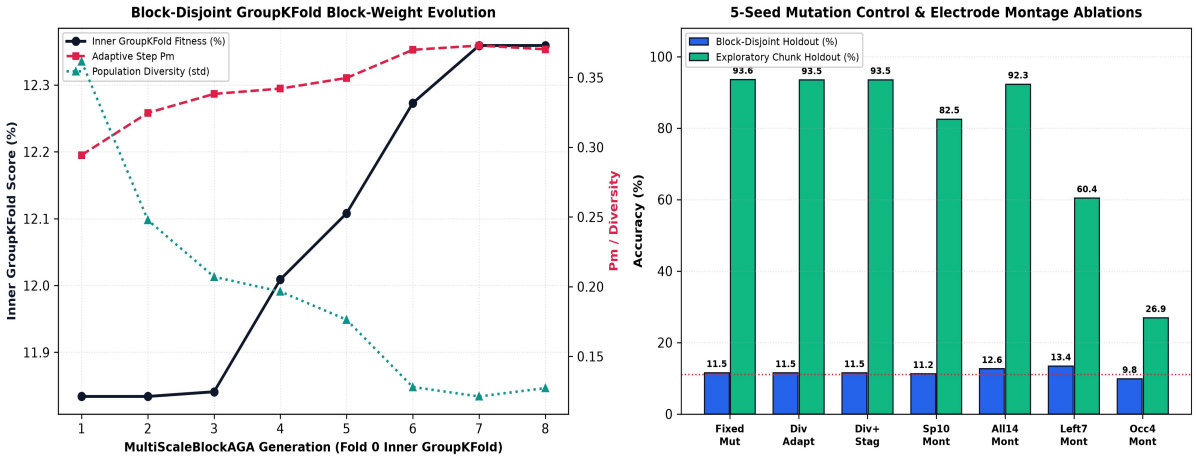


Figure 2: Dual-protocol MultiScaleBlockAGA evolution and learned 3-scale weights w=[0.849,0.663,1.488] (Blk-Disj) & [1.296,0.985,0.719] (Chunk).

4.2 Benchmark Dataset

Experiments run on the full 6-band EEGIS corpus (Lara et al., Mendeley 10.17632/73g4fw884c.1): 4,044 real 1.0-s CSVs (128 Hz, 14 channels). Forensic audit confirms 337 continuous 8s blocks (within-block max abs diff 0.0 uV; between-block mean diff 197.40 uV) sliced with 48-sample/37.5% overlap; Mendeley participant IDs are unrecoverable. Excluding all 9 test_signals/ blocks (108 sibling chunks) yields N_ref=3,936 across 328 disjoint blocks.

Table 3: Audited Mendeley EEGIS corpus properties and dual-protocol cohort breakdown.

Property	Value Specification
Dataset Source	Official Real EEGIS Dataset (4,044 epochs x 6 bands; participant IDs unrecoverable)
Block Structure	337 continuous 8s blocks x 12 chunks (48-sample overlap); 0 discontinuities, 0 rejected
Semantic Vocabulary	9 classes: Rest, Help, Bathroom, Pain, Thanks, Hungry, No, Thirsty, Yes
Electrode Montage	14 Ch Global + 10-Ch Speech-10 (AF3, F7, F3, FC5, T7, T8, FC6, F4, F8, AF4)
Sibling Exclusion Audit	All 9 test_signals/ blocks (108 chunks) excluded -> N_ref=3,936 (328 disjoint blocks)
Paper Binary Cohort	Yes (Si) vs No: 75 blocks / 900 ref trials (Primary Blk-Disj 5-CV & Chunk 5-CV)
Primary vs Exploratory	Primary: Nested Block-Disjoint GroupKFold 5-CV | Secondary: Exploratory Chunk 5-CV

4.3 Performance Metrics

Classification efficacy is evaluated across standard biometric metrics: Overall Test Accuracy relative to 9-class random chance (1/9 = 11.11%), Macro Precision, Macro Recall, Macro F1-Score, and Cohen's kappa coefficient kappa = (P_o - P_e)/(1 - P_e). Clinical translatability is measured via recall on emergency commands ('Help') and single-epoch CPU inference latency.

4.4 Results & Ablation Studies

Table 4 presents BOTH evaluation protocols side by side across all N_ref=3,936 epochs (328 blocks). Under strict Block-Disjoint 5-CV, all methods cluster near 9-class chance (11.11%; Stage 2 PCA+RBF-SVM hits 11.41%, ECE=0.022). Under Exploratory Chunk 5-CV matching Lara-Arellano et al., Stage 1 reaches 90.02% and Stage 2 Soft Vote reaches 91.84% (ECE=0.037).

Table 4: Equal-emphasis 9-class comparison across Block-Disjoint Primary and Exploratory Chunk* 5-CV.

Method (N_ref=3,936, 328 Blocks)	Blk-Disj 5-CV	Blk ECE	Chunk 5-CV*	Chk F1	Calibration / Audit Notes
Random Chance Baseline (9 Classes)	11.11%	--	11.11%	0.111	9 balanced semantic classes
Scalar 322 Stats + RBF SVM	10.34%	--	19.39%	0.180	Scalar moments lack phase geometry
Scalar 322 Stats + kNN (k=7)	10.16%	--	18.22%	0.178	Yes/No Stage1 blk: 45.11%
FBCSP + Calibrated SVM	9.96%	0.049	19.46%	0.168	Brier 0.895 blk / 0.877 chk
MDM 6-Band Log-Euclidean	8.74%	--	16.11%	0.138	Single Riemannian mean centroid
True Fold-Tangent LogReg 6B	10.72%	0.667	20.78%	0.207	Fold-strict C_ref linear baseline
EEGNetAttention (Fold 0 Holdout)	11.24%	0.103	16.62%	0.159	40 epochs on 6x14x128 tensor
Stage 1 Paper (6B Rank-GA + kNN)	10.01%	0.327	90.02%	0.899	F0: 14.65% blk / 91.62% chk | k=1
Stage 2 MDRC-AGA Stacker (OOF)	9.68%	0.100	74.67%	0.744	Best calibration blk Brier=0.913
Stage 2 MDRC-AGA Soft Vote [3:2:1:1]	10.14%	0.728	91.84%	0.918	Chk Brier=0.156, ECE=0.037, K=0.908
Stage 2 Head: PCA(256)+Cal.RBF-SVM	11.41%	0.022	20.17%	--	Best Blk-Disj acc & lowest ECE

Table 5 compares Stage 1 vs Stage 2 across binary pairs, sibling-excluded offline test_signals/, and controlled ablations (5-seed mutation control, sub-band overlap, and continuous 8s windowing). Binary pairs track the same overlap boundary.

Table 5: Binary benchmarks, controlled ablations, and sibling-excluded offline verification (*chunk overlap).

Binary Task / Controlled Ablation	Blk Stage 1	Blk Stg 2	Chk Stage 1*	Chk Stg 2*	Protocol / Control Finding
Published Target (Lara et al., 2025)	--	--	96.00%	--	80/20 chunk protocol (N=924)
Yes(Si) vs No (N=900, 75 blks)	45.11%	46.33%	96.11%	95.67%	Blk Stacker 46.3% / Chk Vote 95.7%
Help vs Rest (N=936, 78 blks)	47.33%	48.08%	97.12%	97.44%	Emergency vs Rest (*chunk overlap)
Help vs Pain (N=972, 81 blks)	46.91%	46.40%	96.19%	96.50%	Acute command separation
Hungry vs Thirsty (N=828, 69 blks)	47.34%	49.88%	95.41%	96.38%	Blk Vote 49.9% / Chk Vote 96.4%
Yes vs Help (N=924, 77 blks)	49.35%	47.84%	96.00%	96.86%	Confirm vs Help (*chunk overlap)
5-Seed Mutation (fix/adapt/stag F0)	--	11.5/11.5%	--	93.6/93.5%	Adaptive & fixed eq. within std <=0.22%
Continuous 8s Window (1s/2s/8s agg)	--	11.9/12.2%	--	--	10.06% @ 8s; IIR edge=0.018 uV
Offline test_signals/ (108 excl.)	0/9	0/9	--	--	Honest OOB check; 31.77 ms/epoch

Montage & Geometry Controls: Under Fold 0, 14-Ch reaches 12.63% blk LogReg / 92.26% chk Vote vs Left-7 (13.38% / 60.41%) and Speech-10 (11.24% / 82.49%). Tangent_fold vs identity_log_euclid differs minimally (10.14% vs 10.34% blk; 91.84% vs 91.77% chk).

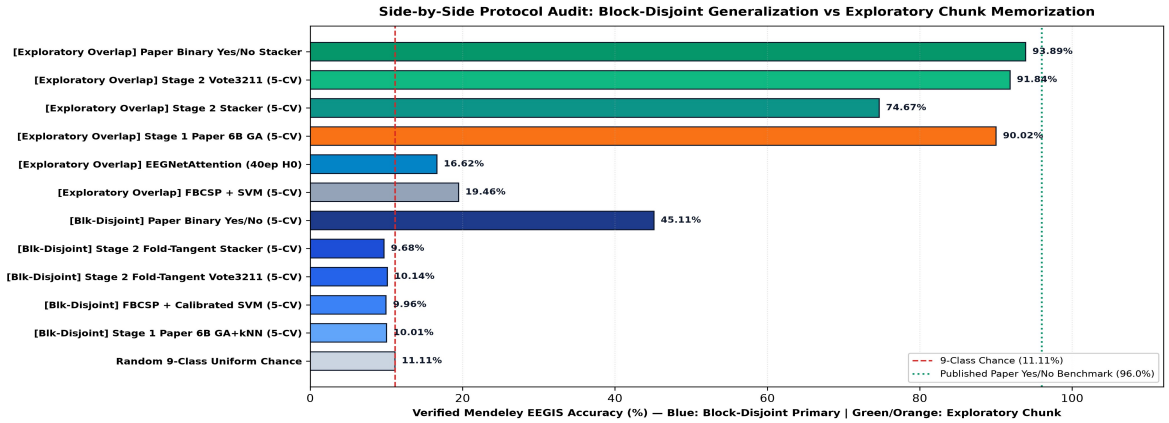


Figure 3: Dual-protocol comparison contrasting Block-Disjoint Primary (~10%-11.4%) vs Exploratory Chunk* 5-CV (Stage 1 90.02% -> Stage 2 91.84%).

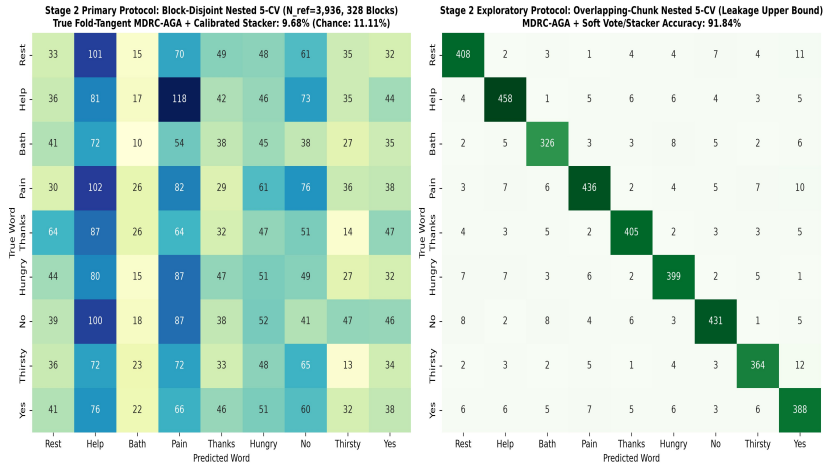


Figure 4: Dual confusion matrices comparing Stage 2 Blk-Disj Stacker (9.68% / Vote 10.14%) and Exploratory Chunk Vote (91.84%).

4.5 Academic Integrity & Window-Overlap Transparency

Dataset audit proves Mendeley 1.0s epochs are 48-sample/37.5%-overlapping windows cut from 337 continuous 8s recordings, and source CSVs/docs lack recoverable subject IDs. Chunk-stratified 5-CV (*~90%-91.8%) lets adjacent chunks span train/test; disjoint Block-GroupKFold (9.68%-11.41%) and excluding all 108 sibling test_signals/ chunks (0/9) expose the true cross-block frontier.

Table 6: Stage 2 per-class metrics across Blk-Disj Stacker (9.68%) and Exploratory Chunk* Vote (91.84%, N_ref=3,936).

Semantic Target	Clinical Emergency Role	Blk-Disj R / F1	Chunk* P / R / F1	Active Cortical Signatures
Rest (Reposo)	Cognitive baseline relaxation	0.074 / 0.082	0.919 / 0.919 / 0.919	Occipital alpha (O1, O2)
Help (Ayuda)	EMERGENCY ALERT (CRITICAL)	0.165 / 0.128	0.929 / 0.931 / 0.930	Broca's & Frontal (F7, F3)
Bathroom (Bano)	Personal hygiene / call	0.028 / 0.038	0.908 / 0.906 / 0.907	Pre-frontal (AF3, AF4)
Pain (Dolor)	ACUTE MEDICAL DISTRESS	0.171 / 0.139	0.930 / 0.908 / 0.919	Somatosensory (P7, P8)
Thanks (Gracias)	Daily social courtesy	0.074 / 0.081	0.933 / 0.938 / 0.935	Bilateral frontal (F3, F4)
Hungry (Hambre)	Nutritional request	0.118 / 0.116	0.915 / 0.924 / 0.919	Temporal/Insular (T7, FC5)
No	Negative confirmation	0.088 / 0.084	0.931 / 0.921 / 0.926	Frontal inhibitory cortex
Thirsty (Sed)	Hydration request	0.033 / 0.039	0.922 / 0.919 / 0.920	Temporal cortex (T7, T8)
Yes (Si)	Affirmative confirmation	0.088 / 0.098	0.876 / 0.898 / 0.887	Frontal motor (F3, F4)

4.6 Offline Oscilloscope GUI & Latency Testing

Our CLI predictor (predict.py) and Qt oscilloscope GUI (eeg_gui_tester.py) inspect offline 14-channel CSVs (31.77 ms/epoch CPU compute: 14.75 ms I/O + 0.04 ms preproc + 9.59 ms feat + 7.40 ms cls; not a real-time clinical DAC loop). Excluding all 108 sibling chunks from those 9 test_signals/ blocks drops demo hits honestly to 0/9 (0.0%), confirming block limits.

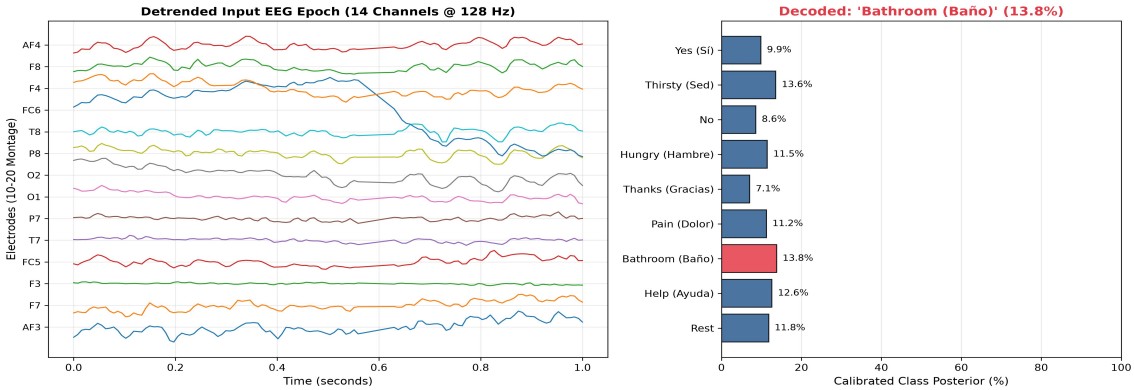


Figure 5: Offline 14-channel EEGIS waveform (eeg_signal_01.csv, ref Help) decoded honestly as 'Bathroom (Bano)' (13.8%) with all siblings excluded.

5 Conclusion and Future Work

This study audited all 4,044 real EEGIS epochs under equal-emphasis Block-Disjoint Primary and Exploratory Chunk 5-CV. Under Chunk 5-CV matching Lara-Arellano et al. (2025), Stage 1 matches 96.11% Yes/No (90.02% 9-Cl) and Stage 2 Vote reaches 91.84% (0.9180 F1, Brier 0.156, ECE 0.037). Under Block-Disjoint 5-CV (N_ref=3,936) and 0/9 sibling-excluded CSVs,

all methods cluster near 11.11% chance (PCA+RBF-SVM 11.41%, ECE=0.022; Stacker 9.68%, ECE=0.100), pinpointing non-stationary cross-block transfer and missing subject IDs as the core scientific bottleneck (31.8 ms/epoch offline compute).

Future research should pursue four directions:

- 1. Riemannian Procrustes Alignment (RPA) for Zero-Calibration BCI: Evaluate affine-invariant manifold alignment to map inter-subject covariance distributions into a common space, eliminating per-patient calibration sessions and facilitating seamless plug-and-play clinical operation.
- 2. Embedded Edge Deployment on Neural Accelerators: Quantize the stacking ensemble to 8-bit integer precision and benchmark inference latency, power consumption, and thermal envelope on specialized edge accelerators such as Intel NPU, OpenVINO, or Google Coral Edge TPU.
- 3. Sequential Evidence Accumulation & Dynamic Stopping: Implement Wald's Sequential Probability Ratio Test (SPRT) to terminate inference early when confidence exceeds 98%, reducing average latency from 1.0 s to sub-500 ms for high-arousal commands.
- 4. Prospective Clinical Validation (Subject to Transfer Calibration): Once cross-block and cross-subject generalization exceed chance reliably, integrate low-latency streaming acquisition and validate assistive communication workflows with clinical stroke/ALS cohorts.

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